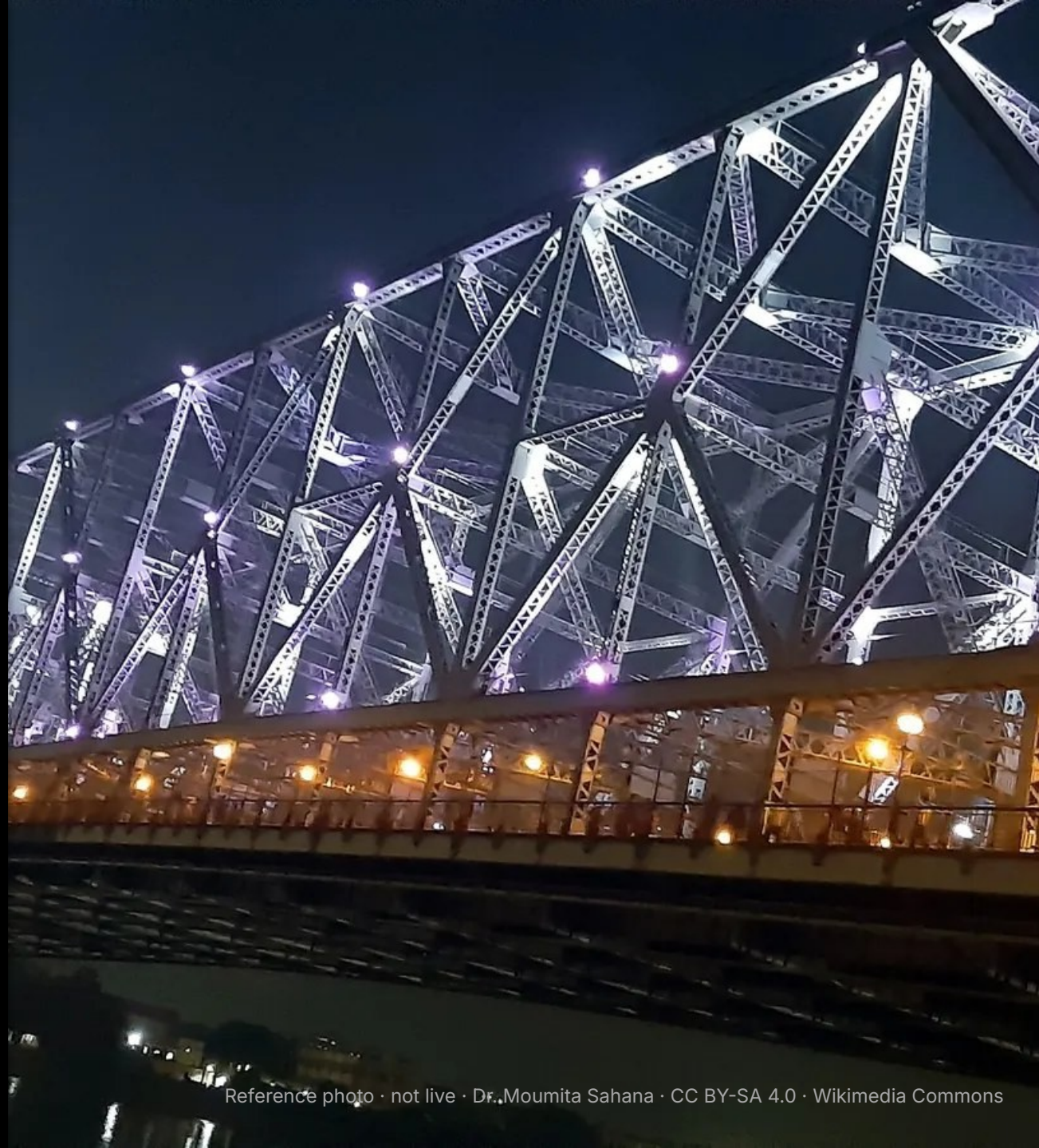
Sahoshi | সাহসী

থামিয়ে নয়, সামলে।

Keep Kolkata moving.

One live picture of Kolkata's roads, rain and crowds for the traffic police, KMC, disaster response and ambulance (108) crews. The AI proposes; a named officer decides.

Open source (MIT) · runs on one on-premises AI computer · github.com/shubro18202758/sahoshi



Reference photo · not live · Dr. Moumita Sahana · CC BY-SA 4.0 · Wikimedia Commons

THE PROBLEM

Kolkata loses its roads to the same few shocks, every year

251.6 mm

of rain in 24 h on 22–23 Sep 2025 — the heaviest in about four decades; roads went under within the hour, 8–12 people died, most by electrocution in waterlogged streets (IMD; Reuters/PTI).

High tide

closes the outfall gates on the Hooghly — the same rain waterlogs far faster when it meets a high tide.

10,000+

police a day on Durga Puja duty, about 30% on traffic, with ad-hoc one-ways and goods-vehicle bans (Kolkata Police, 2025).

≈ 1 lakh

vehicles a day on Howrah Bridge alone; one breakdown cascades across both banks.

15–45 min

corridor freezes for VIP movements between Nabanna, the city and the airport.

One picture missing

traffic police, KMC, WBSDMA, the power utility and ambulance (108) each hold a piece — coordination runs on phone and radio.

See · Decide · Act — one shared, Bengali-first picture

See

Roads, live traffic, rain, the Hooghly tide, pumps, IMD/NDMA alerts, Bengali and English news, field reports spoken in Bengali, camera road state — on one map.

Look ahead

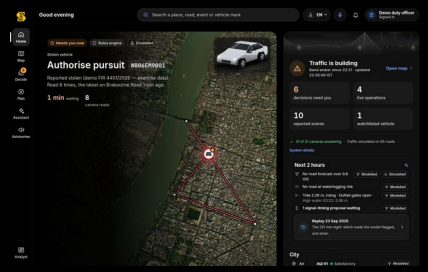
Corridor congestion forecast 15–120 min ahead with uncertainty; a waterlogging nowcast ranks every road by risk with its lead time.

Decide

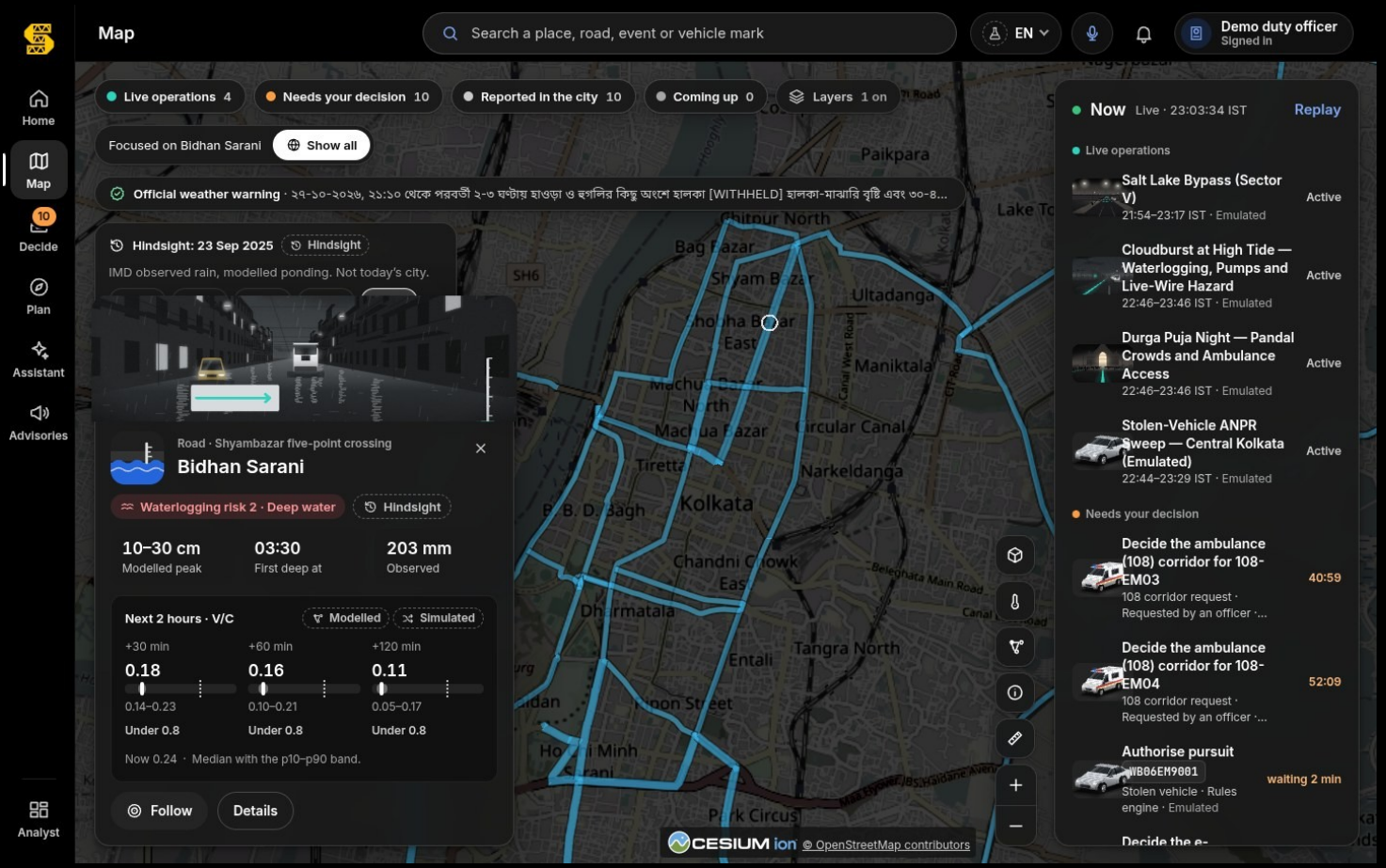
One proposal at a time with what changes if approved: a diversion, a signal-timing change tested in a traffic twin, a rolling hold instead of a freeze, an ambulance window, a live-wire hazard request.

Act

A named officer approves or rejects — nothing acts on its own; no answer, the plan holds. Advisories go out in বাংলা, English and हिंदी, read aloud.



Waterlogging risk and the next two hours, on every road



Rain × tide × terrain × pumps

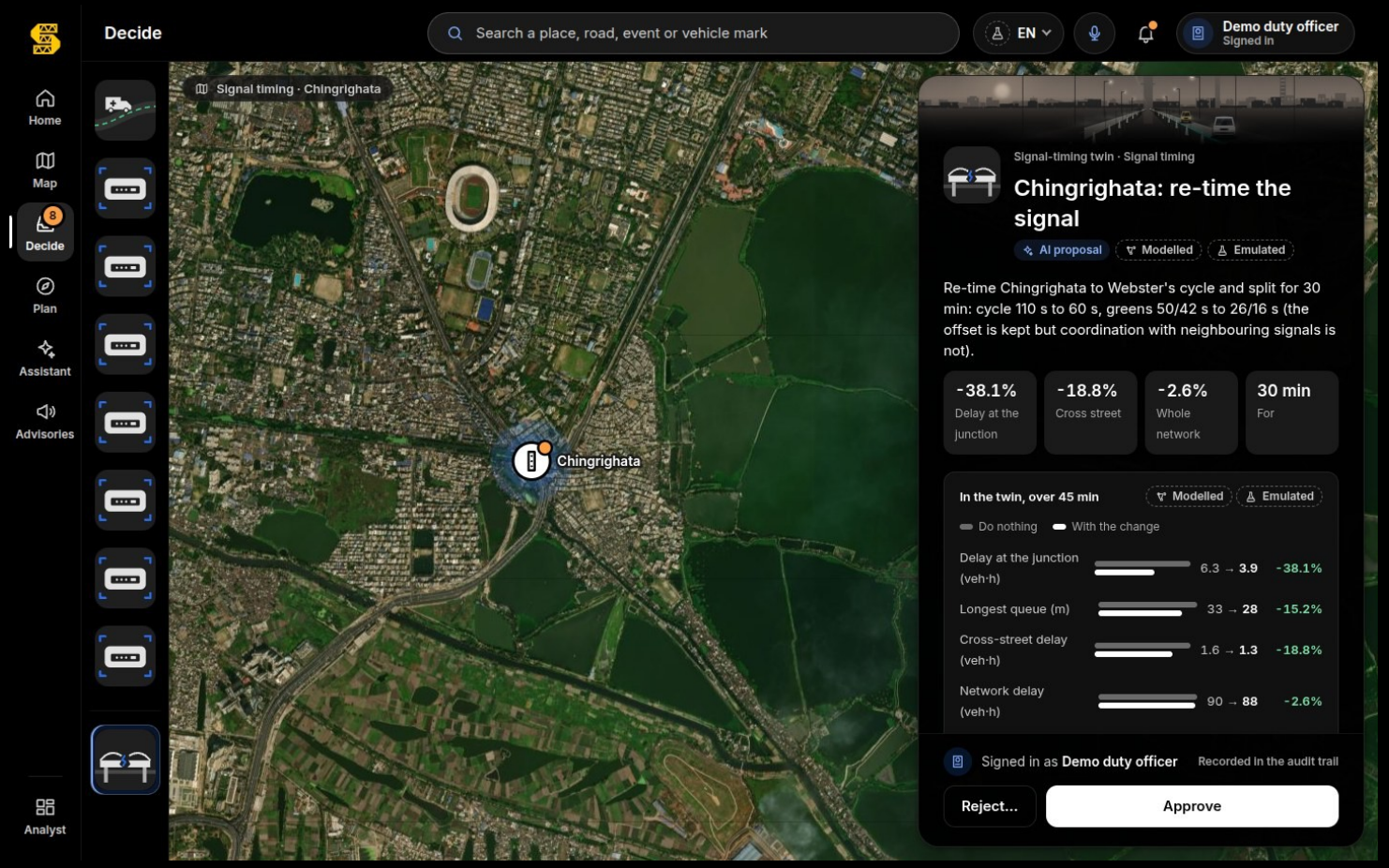
Rain: Open-Meteo, LIVE. Tide: Hooghly model at Garden Reach, MODELLED. Terrain: Copernicus 30 m DEM and OSM drains. Pumps: KMC FFEWS adapter, EMULATED until access.

Risk class 0–3 per road, depth band, lead time to deep water — and the drivers behind it, each labelled.

Forecast

Chronos-2 forecasts every corridor 15–120 min ahead with p10–p90 bands; crossings of 0.8 V/C become diversion proposals before the queue forms. SIMULATED traffic until Kolkata traffic is recorded.

The AI proposes. A named officer decides.



Tested before it is shown

Every signal-timing proposal is simulated in a cell-transmission traffic twin of Kolkata against doing nothing: delay, queues, cross streets, the whole network. It is offered only if it helps without moving the jam.

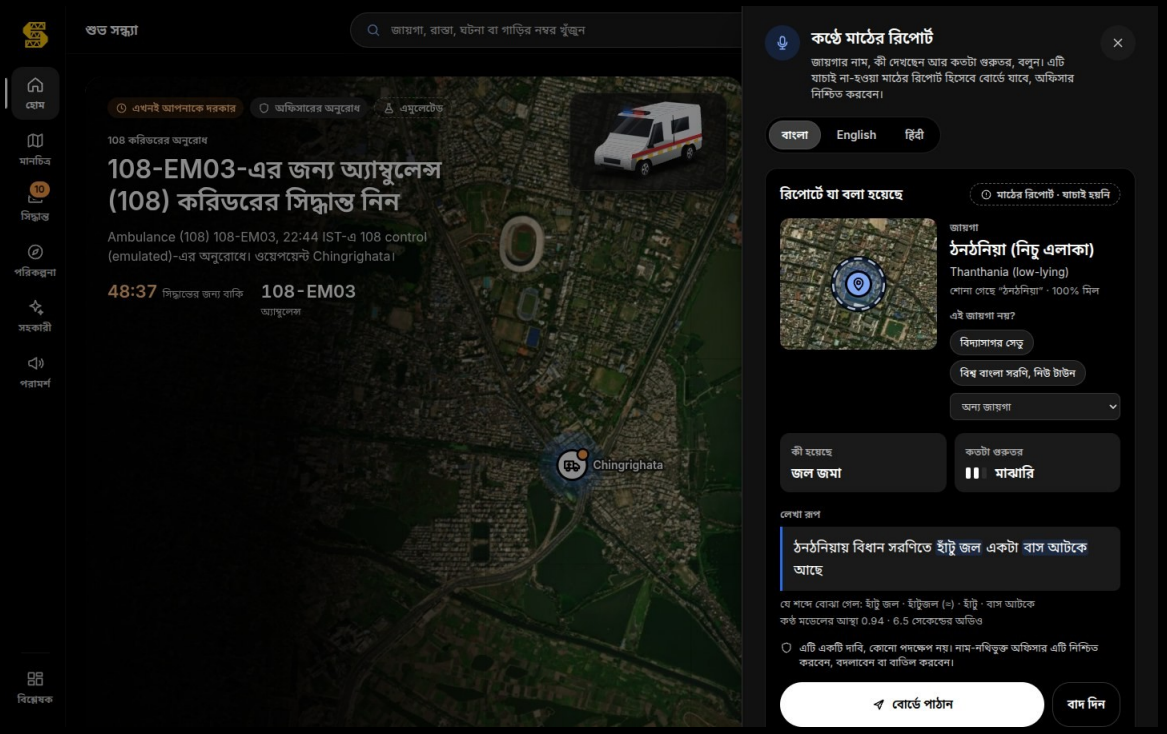
Accountable

Evidence, what changes, confidence and assumptions on every card. Approve or reject is recorded with the officer's name. No answer: the current plan holds.

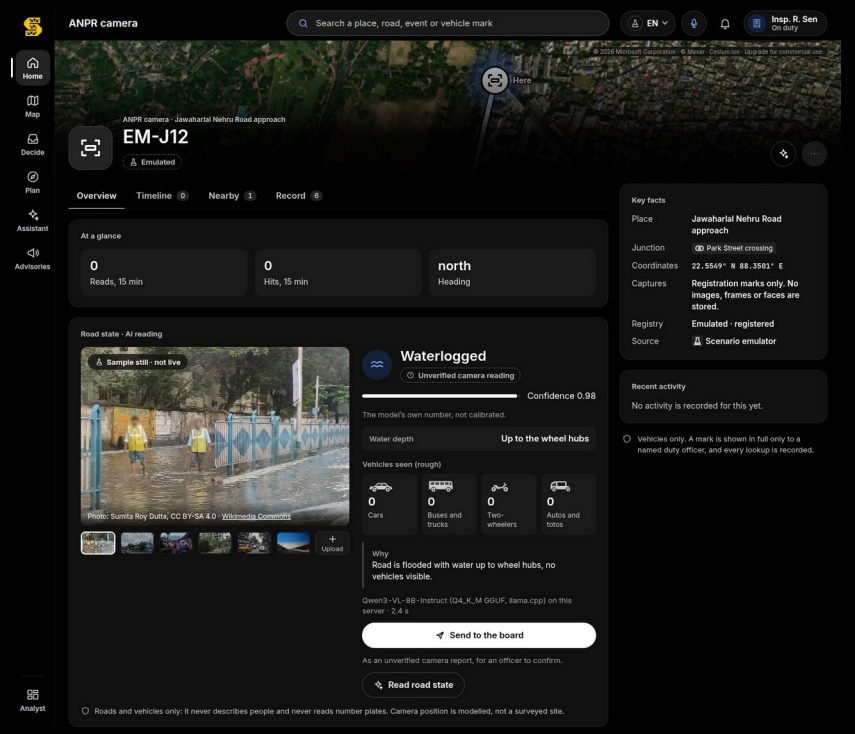
Explained in two languages

A local language model writes the rationale in English and Bengali; any number not in the twin's metrics is rejected.

Bengali voice reports and camera road state

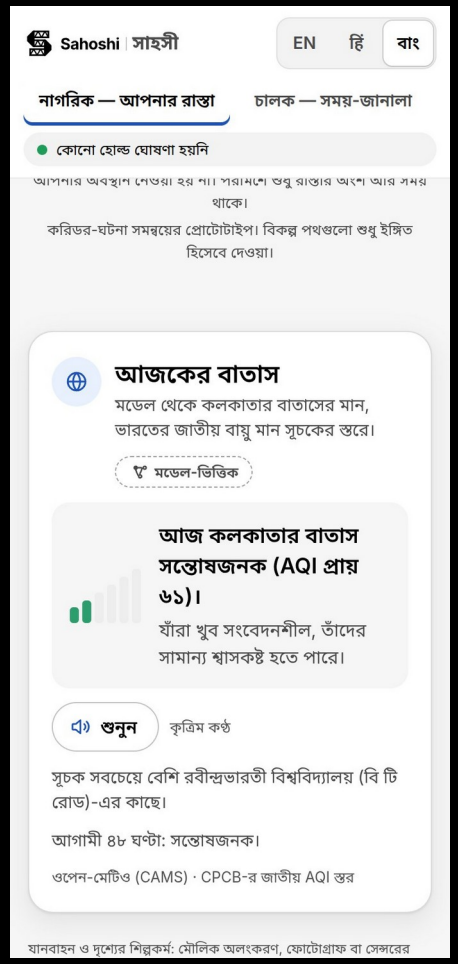


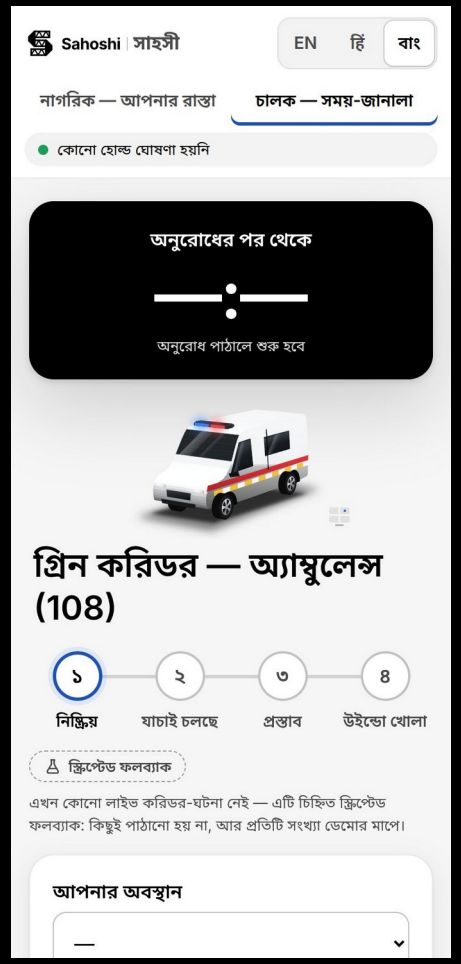
An officer speaks in Bengali; the on-premises speech model returns place · kind · severity with the transcript — an unverified field report for a named officer to confirm. Audio is not stored.



Qwen3-VL reads waterlogged / blocked / tree-fall / stalled bus / clear — roads and vehicles only. It refuses any question about people and never reads plates.

Check your road — in বাংলা, read aloud





Plain words, own language

Waterlogging on your road, the tide and the rain, holds and diversions — in বাংলা, English and हिंदी, with a Listen button (synthetic voice).

Honest by design

Anything simulated or emulated says so; an officer's unverified report never reaches the public page.

Ambulance (108) crews

A one-tap green-corridor request that a named duty officer approves.

OPEN, ON-PREMISES AI

2025–26 open-source models, all on one computer in the control room

Forecast	Chronos-2 (Apache-2.0)	Zero-shot corridor forecasts; error 55% below the seasonal baseline in backtest (SIMULATED history)
Waterlogging	Rain × tide × DEM × pumps	Replay of 22–23 Sep 2025 on IMD-observed rain: 13 of 13 reported places flagged
Signal twin	Cell-transmission model + Webster / max-pressure	Shockwave speed within 0.2% of theory; a 30-min city run in 0.12 s
Speech	Bengali Whisper (Apache-2.0), Svara-TTS	11.5% word error, place matched 53/60 (synthetic round trip)
Vision	Qwen3-VL-8B (Apache-2.0)	11 of 12 road states right; 9 of 9 refusal probes refused
Assistant	Local Gemma-class LLM + retrieval	Answers grounded in the live picture; numbers checked against tool results

No data leaves the control room; it keeps working when the internet fails in a cyclone. No licence fees.

EXPECTED IMPACT · MODELLED

Delay grows with the square of a freeze — so rolling holds change everything

100×

less junction-freeze delay (central case; range 25×–506×)

357,188 → 3,572

commuter-minutes per VIP movement on the Nabanna → airport corridor (13 junctions, 29 km)

≈ 33,000

golden-hour minutes a year returned to ambulance (108) trips

45–120 min

warning before a road goes under (2025 replay, IMD rain)

Model outputs on Kolkata's OpenStreetMap network under stated assumptions (IMPACT_REPORT.md) — not field measurements, and never a claim about lives saved.

GUARDRAILS

Built to strengthen public institutions — safely

Vehicles only

No faces, no persons, no tracking of individuals. Number plates stay masked until a named duty officer is on duty.

Human in charge

The AI only proposes. A named officer approves every action; with no answer, the current plan holds. Every decision is audited.

Honest labels

LIVE, MODELLED, SIMULATED, EMULATED — on every number and every feed that is not yet connected.

Events, not people

News and social signals are aggregate, about places and situations; posters are never stored or scored.

Fair enforcement

No e-challan can be approved against a theft victim; penalties are recommendations a human reviews.

Data stays home

On-premises, retention limits, in the spirit of the DPDP Act 2023.

IMPLEMENTATION

A 12-month pilot through one monsoon and one Durga Puja

Month 0–1	Read-only data MoU: Kolkata Traffic Police, KMC (FFEWS), WBSDMA, ambulance (108); nodal officers named.
Month 1–2	Install the on-premises unit; City Pack finalised for the pilot division; adapters switched from EMULATED to LIVE.
Month 2–4	Shadow mode: Sahoshi proposes, officers work as usual; weekly comparison.
Month 5–12	Live pilot (e.g. Esplanade – Park Street – Rashbehari – Gariahat) through the monsoon and Puja; officers approve.
Month 12	Independent review; city-wide roll-out over the next year.

Cost

Pilot, 12 months: ≈ ₹45–55 lakh

Two on-premises AI units ₹8–10 lakh · engineering ₹29 lakh · optional sensors ₹4 lakh · training ₹2 lakh · 15% contingency.

City-wide, two years: ≈ ₹2–2.5 crore

Running cost ≈ ₹40–60 lakh a year.

Software licences: ₹0

Open models, OpenStreetMap, Sahoshi under MIT — existing CCTV, control rooms and FFEWS sensors are reused.

SCALABILITY

A city is a folder

The engine — live picture, forecasting, nowcast, voice, vision, decision loop, console — is the same everywhere. A City Pack holds what differs: the OpenStreetMap extract and corridors, gazetteer, agencies and emergency numbers, plate format, UI languages, local scenarios and impact parameters.

The prototype proves it: it was first built for another Indian city, and the same build tools now produce Kolkata; a test builds both.

Per district: about 1–2 weeks of local data work, local adapters (CCTV, signals, flood sensors) and local SOPs. The software, models and hardware stay the same.

Howrah · Bidhannagar

already inside the Kolkata pack

Siliguri

gateway to the North-East; NH10/NH31 choke-points

Asansol–Durgapur

industrial corridor

Patna · Ranchi

Bihar and Jharkhand; Hindi already first-class

EVIDENCE

It runs today

Working prototype

Console, citizen and driver pages, five AI services and the traffic twin, on one on-premises unit.

Real Kolkata roads

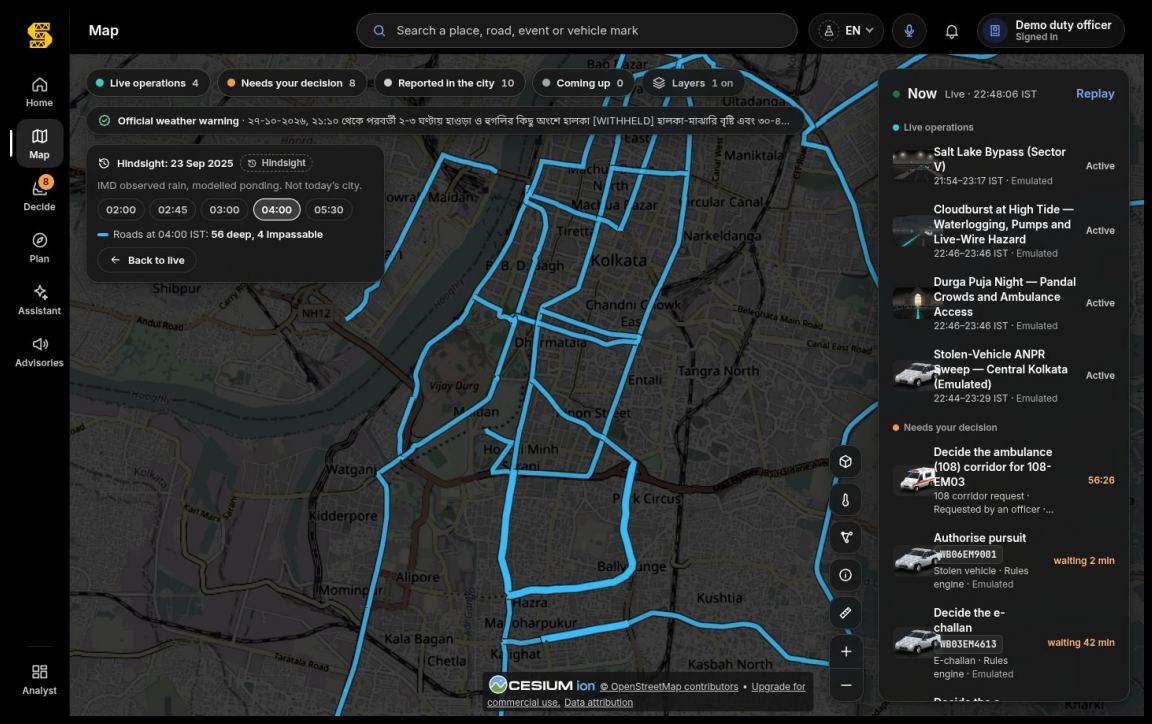
49 junctions, 65 road segments, 125 km from OpenStreetMap; the Nabanna → airport route is 29 km.

Thousands of automated tests

Python suites, the console's unit and browser tests, contract replays and Rust tests — run on every change.

Open source

github.com/shubro18202758/sahoshi — setup, technical design, sample data and test cases.



Replay of 22–23 Sep 2025 on IMD-observed rain (OBSERVED rain, MODELLED ponding): every news-reported waterlogged place on the network was flagged — the night almost the whole city flooded.



Sahoshi

সাহসী

ধন্যবাদ

Thank you

github.com/shubro18202758/sahoshi

Reference photo · not live · Alexey Komarov · CC BY-SA 4.0 · Wikimedia Commons